



COMPUTER SCIENCE AND SOCIETY

AGENDA

Computer Science, Society, and the Information Society

1

The Role of Data in the Information Society

2

Economic Impact of the Information Society

3

Social Impacts of the Information Society

4

Infrastructure Vulnerability, Computer Science and the Military

5

Responsibility in Computer Science

6

UNIT 5

**INFRASTRUCTURE VULNERABILITY/
COMPUTER SCIENCE AND THE MILITARY**



On completion of this unit, you will be able to...

- identify social and technical infrastructures that are worth protecting.
- name and describe the major types of attacks and accidents that threaten critical infrastructure.
- recommend basic steps for IT risk management.
- name factors that are necessary for the implementation of IT security measures.
- identify and describe the connections between computer science and the military.
- understand how the military influence upon civilian life is rooted in technological developments.
- reflect on the ambivalence of technology.



1. Name some critical infrastructures!
2. Explain what is meant by “Cyber War”
3. Give some examples for dual use of technology!

What if the Internet went down for one day?

- **93%** of people in the US use it daily
- Running **business, banking, society, ...**
- **Damages in the billions?** Probably not (Nuwer, 2017)

→ but: vulnerability is a major concern!



What could go wrong?

- Hackers
- Viruses
- Ransomware
- Accidents
- Human Error
- Bugs

...

Technical:

- Electricity
- Oil
- Telecommunications
- Transportation
- Water
- Banks
- ...

Political/ Social

- Government/ administration
- State IT
- Protection from Propaganda/
Misinformation
- ...

Risks are part of life

- manage them!
- analyse risks in time, take **countermeasures**
- Start with simple measures: e.g. regular updates and backups
- give priority to **management of IT security**
- Measures must be supported by **top management** and **employees**



MILITARY AS THE DRIVER OF COMPUTER SCIENCE

- War contradicts the rational world view, **pacifist attitude** of many computer scientists
- **Aggression** as human nature?
- Development of Computer science and ICT was largely **driven by wars / military**
- E.g. **ARPANET**, precursor to the Internet, started as a military project
- E.g. AI: early development by **Alan Turing**



CYBER WAR

- Modern warfare usually includes elements of **Cyber War**, e.g. hacker attacks on infrastructure
- Interconnected ICT systems: high **vulnerability**
- Instruments: e.g. **computer viruses**, market platforms, digital currencies....



- individuals **no longer helpless** in war?
- e.g. track the movements of the enemy to support the defence
- See “**Arab Spring**”, 2010s
- hacker attacks to **paralyze enemy infrastructure**

Same technology can be used for **civilian** or **military** purposes:

- **Drones**
- **Nuclear power**
- Much of ICT
 - E.g. Sony **Hololens 2**

→ Potential **ethical and legal issues**





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SESSION 3

TRANSFER TASK

The Real Story of Stuxnet

Stuxnet is a piece of **malware** discovered in 2010 that had apparently been developed to damage the nuclear program of Iran. It is still considered one of the **most elaborate malware attacks** ever. It is documented in

- Kushner, D. (2013). The real story of Stuxnet. *IEEE Spectrum*, 50(3), 48–53.

<https://doi.org/10.1109/MSPEC.2013.6471059>



TRANSFER TASK

Read the case study in order to answer the following questions:

1. Briefly describe how Stuxnet worked and how it was able to **gain access** to restricted systems.
2. Knowing this, which **policies** would you implement to protect vulnerable **critical infrastructures**? How would you implement them?
3. What are typical **threat scenarios** for the organization you work in? Which of the advice from 2. can be useful for that?

TRANSFER TASK
PRESENTATION OF THE RESULTS

Please present your
results.

The results will be
discussed in plenary.





1. What is **not** a common kind of attack/ accident?
 - a) Viruses
 - b) Bugs
 - c) Bacteria
 - d) Human error



1. What is **not** a critical infrastructure?
 - a) Air
 - b) Oil
 - c) Water
 - d) Banks



3. Which technology was originally developed by the **military**?

- a) Smartphone
- b) Internet
- c) Mouse
- d) Hard disks

LIST OF SOURCES

Kleppmann, M. (2017): *Designing data-intensive applications. The big ideas behind reliable, scalable, and maintainable systems*. 1st Edition. Sebastopol, CA: O'Reilly.

Nuwer, R. (2017, 7. Februar). What if the internet stopped working for a day? BBC Future. <https://www.bbc.com/future/article/20170207-what-if-the-internet-stopped-for-a-day>

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